

Drixal Service Platform

Product, Functional and Technical Specification

Everything the product owner needs to know about the current platform

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Purpose: this document describes what Drixal does today, how it is organised, which roles and workflows exist, what the underlying rules and status lifecycles are, and what security and UX guarantees it provides. It is written for the product owner, but also serves developers, QA, and business stakeholders.

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1 Executive Summary

Drixal is a multi-tenant service-management platform that connects service discovery, customer requests, provider review, order conversion, assignment, scheduling, and customer tracking in one role-aware workflow.

The product is delivered as a modern web application (Nuxt 4, TypeScript, MongoDB). It currently serves four role families:

- **Super Administrators** govern provider-company approval and account status.
- **Company managers** run daily service operations: requests, orders, schedules, customers, services, and team workload.
- **Company employees** (technicians and viewers) review assigned work and coordinate delivery.
- **Customers** discover services, submit requests, and track their own requests and orders in a private *Personal* workspace.

Success for every role means: understand the current operational state and complete the next task with minimal navigation and ambiguity.

2 Product Positioning and Operating Context

Positioning

Drixal carries a service request from public discovery through tenant-controlled approval, conversion, assignment, and private customer tracking in one role-aware workflow.

The primary operating scene is a company manager repeatedly scanning new demand, active work, schedule pressure, customer context, and exceptions throughout the workday. Structured lists, tables, statuses, filters, and contextual actions are the core working materials. The public marketplace and authentication surfaces support entry into this operational system.

2.1 Key product principles

1. The next operational decision matters more than decorative metrics.
2. Role clarity, tenant boundaries, and trustworthy status communication are preserved.
3. Structured business data stays scannable; actions stay contextual.
4. Arabic (RTL) and English (LTR) are equally complete — neither is a fallback.
5. Simple, consistent workflows are preferred over feature-shaped visual novelty.

3 Roles, Workspaces, and Access

The platform uses a single global identity model (User). Company authorization is **contextual**: a user may hold a different role in each company and can hold no company membership at all (a pure customer).

Workspace	Landing route	Who uses it
Personal	/customer	Every user: requests, orders, marketplace, company creation
Company Administration	/company-admin	OWNER, ADMIN, MANAGER roles of the selected company
Employee	/employee	TECHNICIAN, VIEWER roles of the selected company
Platform Administration	/super-admin	Users with the SUPER_ADMIN platform role

3.1 Company roles and permissions

Role	Permissions within the selected company
OWNER	Full company permission set: read company, manage/publish services, decide and convert requests, read/manage customers, read/manage orders
ADMIN	Same full company permission set as OWNER
MANAGER	Same full company permission set as OWNER/ADMIN
TECHNICIAN	Read company, read orders (read-only operational views)
VIEWER	Read company, read services, read requests, read customers, read orders (read-only everywhere)

Platform-level permissions are independent of company memberships:

Platform role	Platform permission
USER	None (can still hold company memberships)
SUPER_ADMIN	Review companies (approve, reject, suspend, reactivate)

Permissions are enforced on the **server**. Hiding a button in the UI is never the only safeguard. Every protected endpoint checks authentication, active membership, role permission, entity ownership, and tenant scope.

4 Functional Modules

4.1 1. Identity, Authentication, and Workspaces

- Account registration with name, email, and password; passwords are hashed with **Argon2id**.
- Sign in / sign out with database-backed, HTTP-only cookie sessions.
- Explicit session DTO exposing the current user, active workspace, company, membership, and permission list.
- A workspace switcher lets a user move between *Personal*, each *company* membership, and *Platform Administration*.
- Same-origin mutation protection (CSRF defence) for all non-GET requests.

- Login rate limiting.
- The selected workspace is persisted per session; a stale or removed company membership falls back safely to Personal.

4.2 2. Public Marketplace

- Public, unauthenticated browsing of the marketplace.
- Listing of eligible services only (see publication rules below).
- Search by service name/description, filters by category, city, and price range (min/max).
- Filter state lives in the URL query string (shareable, refresh-safe, back-button friendly).
- Public company profile pages listing the company's published services.
- Public service detail pages showing pricing, duration, location type, scheduling requirement, category, and provider profile.
- Debounced search input; pagination; loading, error, and empty states.

4.3 3. Service Catalog and Publication Lifecycle

A *service* belongs to exactly one company (the tenant). Each service carries:

- Name, slug, description.
- Category (platform-managed).
- Pricing: FIXED, HOURLY, or CUSTOM (quote).
- Duration in minutes.
- Location type: PROVIDER, CUSTOMER, REMOTE, FLEXIBLE.
- Scheduling requirement (boolean).
- Operational status: ACTIVE / INACTIVE.
- Publication status: DRAFT / PUBLISHED / UNPUBLISHED.

Manager actions: create, edit, publish, and unpublish. New services start as DRAFT and remain invisible to the public until published.

4.3.1 Publication rules (enforced server-side)

A service is publicly visible in the marketplace **only if all** of the following hold:

1. `service.publicationStatus = PUBLISHED`
2. `service.operationalStatus = ACTIVE`
3. `company.status = APPROVED`

4.4 4. Service Requests

Customers submit a request against a published service from the public detail page (login required). A request records customer contact details, an optional preferred date, and a message.

Request status lifecycle:

Status	Meaning
NEW	Submitted, awaiting provider review
UNDER_REVIEW	Provider is reviewing
CONTACTED	Provider contacted the customer
APPROVED	Provider approved; eligible for conversion
REJECTED	Provider rejected the request
CONVERTED	Converted into a service order
CANCELLED	Cancelled (by customer)
CLOSED	Closed

Provider (manager) actions: **approve**, **reject** (with confirmation), and **convert** an approved request into a service order. The server rejects decisions on requests that are no longer decidable and prevents duplicate conversion.

4.5 5. Service Orders and Service Lines

A service order is the execution container. Orders carry:

- Order number (unique per company, e.g. S0-1001), title, description.
- Customer, linked service, optional source request.
- Priority: LOW, MEDIUM, HIGH, CRITICAL.
- Status (see lifecycle below).
- Scheduled date.
- Assigned technician (person name and linked user).
- Embedded *service lines*: title, quantity, assignee, status, cost.

Order status lifecycle:

Status	Meaning
DRAFT	Created, not yet scheduled/assigned
SCHEDULED	A date has been scheduled
ASSIGNED	Assigned to a technician
IN_PROGRESS	Work has started
ON_HOLD	Temporarily blocked/paused
COMPLETED	Finished
CANCELLED	Cancelled by a manager

Valid transitions (enforced server-side): DRAFT → SCHEDULED/CANCELLED; SCHEDULED → ASSIGNED/IN_PROGRESS; ASSIGNED → IN_PROGRESS/CANCELLED; IN_PROGRESS → ON_HOLD/COMPLETED/CANCELLED; ON_HOLD → IN_PROGRESS/CANCELLED.

Manager actions: create orders, add service lines, update line assignment/status, assign an order to a company member, and schedule.

4.6 6. Scheduling and Assignment

- Schedule views (manager and employee) list active orders with dates, customers, services, assignees, and statuses.

- Assignments can target the order level (a company member) and individual service lines.
- Schedule views filter out COMPLETED and CANCELLED orders.

4.7 7. Customers Directory

- Company-scoped customer records (name, phone, email, city, status).
- Customers are auto-created/updated from public request submissions (linked to the requesting user where applicable).
- Searchable, paginated directory; unique phone per company.
- A customer is **not** a company membership — the concepts are separate.

4.8 8. Company Onboarding and Approval

Company status lifecycle:

Status	Meaning
PENDING	Registered, awaiting platform review
APPROVED	Eligible to publish services and appear in the marketplace
REJECTED	Application rejected
SUSPENDED	Previously approved but temporarily disabled

Valid transitions: PENDING → APPROVED/REJECTED; APPROVED → SUSPENDED/REJECTED; REJECTED → APPROVED; SUSPENDED → APPROVED/REJECTED.

- Any user can register a company; an OWNER membership is created immediately and the company starts PENDING.
- Super Admin reviews companies from the platform workspace (search + status filter + pagination).
- Approve, reject (with confirmation), suspend (with confirmation), and reactivate.
- A company that is not APPROVED cannot publish services and does not appear in the marketplace; its own profile screen explains why.

4.9 9. Dashboards

4.9.1 Customer dashboard

Shows the next scheduled service, request/order metrics, recent orders, a request-journey panel, and the user's companies. Quick actions to the marketplace and company creation.

4.9.2 Company Admin dashboard

Shows metrics (services, open requests, active orders, customers), a work-queue table of active orders, and a demand pipeline of request statuses. Quick actions to schedule and service creation.

4.9.3 Employee dashboard

Shows active-order and scheduled-work metrics, a work queue, and a status distribution. Quick action to the schedule.

4.9.4 Super Admin dashboard

Shows platform metrics (total/pending/approved/suspended companies) and a review queue, plus a status distribution panel.

5 User Journeys

Customer

Browse → service details → sign in/register → submit request → provider review → (approved request converts to order) → execution → completion. The customer tracks requests and orders in the Personal workspace.

Company Manager

Review requests → approve or reject → convert → assign employee → schedule → monitor → close. Also maintains services, customers, and the company profile.

Employee

Select company → review assigned work → execute. Read-only for technicians; fully read-only for viewers.

Company Owner

Create company → await approval → configure services → publish → operate. Creating another company remains available from Personal even when the user already has memberships.

Super Admin

Review company context → approve, reject, suspend, or reactivate.

6 Information Architecture and Navigation

Context	Navigation items
Personal	Overview, Requests, Orders, Marketplace, (Create Company)
Company management	Overview, Requests, Orders, Schedule, Customers, Services, Company
Employee	Overview, Orders, Schedule
Platform administration	Overview, Companies
Public	Marketplace, Sign in, Register

Filtering, search, and pagination state are reflected in URLs (e.g. /company-admin/orders?status=IN_PROGRESS).

7 Status and Error UX

- Statuses are shown as consistent badges with text labels; color is never the only status signal.
- Loading, error, and empty states are distinct and actionable.
- Destructive actions (reject, suspend, unpublish, cancel) require an explicit confirmation dialog naming the affected entity.
- Buttons prevent duplicate submissions while processing; forms show field-level errors and preserve valid data.

- The UI adapts to desktop, tablet, and mobile; operational tables degrade to readable layouts on narrow screens.

8 Accessibility and Bilingual Support

- Keyboard navigation, visible focus states, semantic labels, and table headers are required.
- Sufficient contrast and non-color status cues.
- Buttons use button elements; links use links; icon-only controls carry accessible labels.
- Full bidirectional (RTL/LTR) support with the Cairo typeface; hierarchy, icon meaning, and action order are preserved in both directions.

9 Security and Tenant Isolation

- Argon2id password hashing; HTTP-only, same-site cookie sessions; 14-day session lifetime with periodic activity updates.
- Same-origin (CSRF) checks on all mutating requests.
- Role-based access control enforced server-side on every protected endpoint.
- Every business document carries `companyId` and all queries enforce company boundaries.
- Explicit customer DTOs: customers only see their own requests/orders and never internal costs, notes, or identifiers.
- Rate limiting on authentication endpoints.
- Audit logging of significant actions (company decisions, service publication, request conversion, order operations).
- Seed data is for demos only and must never be presented as production business evidence.

10 Technical Overview

- **Stack:** Nuxt 4, Vue 3, TypeScript, Nitro, Tailwind CSS, Nuxt UI, MongoDB, Mongoose.
- **Architecture:** modular monolith; standalone-first, integration-ready. The platform runs completely without an external ERP; future integrations are adapter-based.
- **Data model (current):** users, companies, company memberships, service categories, services, customers, service requests, service orders, auth sessions, audit logs.
- **Indexing:** tenant-scoped operational indexes (e.g. `{companyId, status}`), unique company+slug for services, unique company+phone for customers, unique order numbers, unique membership identity per company.
- **Validation:** server-side validation on all input; explicit command endpoints rather than unrestricted status mutation.
- **Quality:** automated CI runs lint, typecheck, unit tests, and production build on every push/PR.

11 Delivery Status and Roadmap

11.1 Implemented today

- MongoDB/Mongoose models and seed data.
- Authentication (Argon2id, sessions, registration) and contextual workspaces.
- RBAC and role-aware navigation.
- Company registration, onboarding, and super-admin approval/rejection/suspension/reactivation.
- Service catalog CRUD and publication lifecycle.
- Public marketplace with search/filter/pagination and company/service detail pages.
- Customer requests and the approve/reject/convert lifecycle.
- Service orders with service lines, scheduling, and assignment.
- Customer directory and private customer tracking.
- Four role-specific dashboards.
- Audit logging.

11.2 Planned (not yet implemented)

- Full booking lifecycle with appointment slots.
- Service execution checklists, time tracking, and materials.
- Payments and invoices (invoice/payment kept as separate concepts).
- Notifications and notification preferences.
- Reviews and ratings.
- Company subscriptions (platform subscription plans and payments).
- External integrations (ERP adapter, webhooks, sync).

12 Glossary

Term	Definition
Tenant	A company; the boundary for all provider operations and data isolation
Company	A service provider registered on the platform
Service	A concrete offering by a company (e.g. AC Maintenance)
Service request	A customer's request for a service, before an order exists
Service order	The execution container for actual work
Service line	A line item inside an order (a task or sub-service)
Publication status	DRAFT / PUBLISHED / UNPUBLISHED visibility of a service
Operational status	ACTIVE / INACTIVE offering state of a service
Workspace	A role-based application context (Personal, Company, Platform)

Term	Definition
Personal workspace	The customer-facing context every user owns